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► B ► M9 REGULATION (EU) No 575/2013 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL

of 26 June 2013

on prudential requirements for credit institutions and amending Regulation (EU) No 648/2012 ◀

► C2 (Text with EEA relevance) ◀

(OJ L 176, 27.6.2013, p. 1)

Amended by:

		Official Journal		
		No	page	date
► <u>M1</u>	Commission Delegated Regulation (EU) 2015/62 of 10 October 2014	L 11	37	17.1.2015
► <u>M2</u>	Regulation (EU) 2016/1014 of the European Parliament and of the Council of 8 June 2016	L 171	153	29.6.2016
► <u>M3</u>	Commission Delegated Regulation (EU) 2017/2188 of 11 August 2017	L 310	1	25.11.2017
► <u>M4</u>	Regulation (EU) 2017/2395 of the European Parliament and of the Council of 12 December 2017	L 345	27	27.12.2017
► <u>M5</u>	Regulation (EU) 2017/2401 of the European Parliament and of the Council of 12 December 2017	L 347	1	28.12.2017
► <u>M6</u>	Commission Delegated Regulation (EU) 2018/405 of 21 November 2017	L 74	3	16.3.2018
► <u>M7</u>	Regulation (EU) 2019/630 of the European Parliament and of the Council of 17 April 2019	L 111	4	25.4.2019
► <u>M8</u>	Regulation (EU) 2019/876 of the European Parliament and of the Council of 20 May 2019	L 150	1	7.6.2019
► <u>M9</u>	Regulation (EU) 2019/2033 of the European Parliament and of the Council of 27 November 2019	L 314	1	5.12.2019
► <u>M10</u>	Regulation (EU) 2019/2160 of the European Parliament and of the Council of 27 November 2019	L 328	1	18.12.2019
► <u>M11</u>	Regulation (EU) 2020/873 of the European Parliament and of the Council of 24 June 2020	L 204	4	26.6.2020
► <u>M12</u>	Commission Delegated Regulation (EU) 2021/424 of 17 December 2019	L 84	1	11.3.2021
► <u>M13</u>	Regulation (EU) 2021/558 of the European Parliament and of the Council of 31 March 2021	L 116	25	6.4.2021

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- (d) seek to ensure that securitisation positions to which the same risk weight is applied on the basis of the credit assessments of ECAs are subject to equivalent degrees of credit risk.

The EBA shall submit those draft implementing technical standards to the Commission by 1 July 2014.

Power is conferred on the Commission to adopt the implementing technical standards referred to in the first subparagraph of this paragraph in accordance with Article 15 of Regulation (EU) No 1093/2010.

▼C2*CHAPTER 6****Counterparty credit risk*****Section 1****Definitions***Article 271***Determination of the exposure value**

1. An institution shall determine the exposure value of derivative instruments listed in Annex II in accordance with this Chapter.
2. An institution may determine the exposure value of repurchase transactions, securities or commodities lending or borrowing transactions, long settlement transactions and margin lending transactions in accordance with this Chapter instead of making use of Chapter 4.

*Article 272***Definitions**

For the purposes of this Chapter and of Title VI of this Part, the following definitions shall apply:

General terms

- (1) ‘counterparty credit risk’ or ‘CCR’ means the risk that the counterparty to a transaction could default before the final settlement of the transaction's cash flows;

Transaction types

- (2) ‘long settlement transactions’ means transactions where a counterparty undertakes to deliver a security, a commodity, or a foreign exchange amount against cash, other financial instruments, or commodities, or vice versa, at a settlement or delivery date specified by contract that is later than the market standard for this particular type of transaction or five business days after the date on which the institution enters into the transaction, whichever is earlier;

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- (3) ‘margin lending transactions’ means transactions in which an institution extends credit in connection with the purchase, sale, carrying or trading of securities. Margin lending transactions do not include other loans that are secured by collateral in the form of securities;

Netting set, hedging sets, and related terms

- (4) ‘netting set’ means a group of transactions between an institution and a single counterparty that is subject to a legally enforceable bilateral netting arrangement that is recognised under Section 7 and Chapter 4.

Each transaction that is not subject to a legally enforceable bilateral netting arrangement which is recognised under Section 7 shall be treated as its own netting set for the purposes of this Chapter.

Under the Internal Model Method set out in Section 6, all netting sets with a single counterparty may be treated as a single netting set if negative simulated market values of the individual netting sets are set to 0 in the estimation of expected exposure (hereinafter referred to as ‘EE’);

- (5) ‘risk position’ means a risk number that is assigned to a transaction under the Standardised Method set out in Section 5 following a predetermined algorithm;

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- (6) ‘hedging set’ means a group of transactions within a single netting set for which full or partial offsetting is allowed for determining the potential future exposure under the methods set out in Section 3 or 4 of this Chapter;

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- (7) ‘margin agreement’ means an agreement or provisions of an agreement under which one counterparty must supply collateral to a second counterparty when an exposure of that second counterparty to the first counterparty exceeds a specified level;

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- (7a) ‘one way margin agreement’ means a margin agreement under which an institution is required to post variation margin to a counterparty but is not entitled to receive variation margin from that counterparty or vice-versa;

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- (8) ‘margin threshold’ means the largest amount of an exposure that remains outstanding before one party has the right to call for collateral;
- (9) ‘margin period of risk’ means the time period from the most recent exchange of collateral covering a netting set of transactions with a defaulting counterparty until the transactions are closed out and the resulting market risk is re-hedged;

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- (10) ‘effective maturity’ under the Internal Model Method for a netting set with maturity greater than one year means the ratio of the sum of expected exposure over the life of the transactions in the netting set discounted at the risk-free rate of return, divided by the sum of expected exposure over one year in the netting set discounted at the risk-free rate.

This effective maturity may be adjusted to reflect rollover risk by replacing expected exposure with effective expected exposure for forecasting horizons under one year;

- (11) ‘cross-product netting’ means the inclusion of transactions of different product categories within the same netting set pursuant to the cross-product netting rules set out in this Chapter;

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- (12) ‘current market value’ or ‘CMV’ means the net market value of all the transactions within a netting set gross of any collateral held or posted where positive and negative market values are netted in computing the CMV;

- (12a) ‘net independent collateral amount’ or ‘NICA’ means the sum of the volatility-adjusted value of net collateral received or posted, as applicable, to the netting set other than variation margin;

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Distributions

- (13) ‘distribution of market values’ means the forecast of the probability distribution of net market values of transactions within a netting set for a future date (the forecasting horizon), given the realised market value of those transactions at the date of the forecast;
- (14) ‘distribution of exposures’ means the forecast of the probability distribution of market values that is generated by setting forecast instances of negative net market values equal to zero;
- (15) ‘risk-neutral distribution’ means a distribution of market values or exposures over a future time period where the distribution is calculated using market implied values such as implied volatilities;
- (16) ‘actual distribution’ means a distribution of market values or exposures at a future time period where the distribution is calculated using historic or realised values such as volatilities calculated using past price or rate changes;

Exposure measures and adjustments

- (17) ‘current exposure’ means the larger of zero and the market value of a transaction or portfolio of transactions within a netting set with a counterparty that would be lost upon the default of the counterparty, assuming no recovery on the value of those transactions in insolvency or liquidation;
- (18) ‘peak exposure’ means a high percentile of the distribution of exposures at particular future date before the maturity date of the longest transaction in the netting set;

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- (19) ‘expected exposure’ (hereinafter referred to as ‘EE’) means the average of the distribution of exposures at a particular future date before the longest maturity transaction in the netting set matures;
- (20) ‘effective expected exposure at a specific date’ (hereinafter referred to as ‘Effective EE’) means the maximum expected exposure that occurs at that date or any prior date. Alternatively, it may be defined for a specific date as the greater of the expected exposure at that date or the effective expected exposure at any prior date;
- (21) ‘expected positive exposure’ (hereinafter referred to as ‘EPE’) means the weighted average over time of expected exposures, where the weights are the proportion of the entire time period that an individual expected exposure represents.

When calculating the own funds requirement, institutions shall take the average over the first year or, if all the contracts within the netting set mature within less than one year, over the time period until the contract with the longest maturity in the netting set has matured;

- (22) ‘effective expected positive exposure’ (hereinafter referred to as ‘Effective EPE’) means the weighted average of effective expected exposure over the first year of a netting set or, if all the contracts within the netting set mature within less than one year, over the time period of the longest maturity contract in the netting set, where the weights are the proportion of the entire time period that an individual expected exposure represents;

CCR related risks

- (23) ‘rollover risk’ means the amount by which EPE is understated when future transactions with a counterparty are expected to be conducted on an ongoing basis.

The additional exposure generated by those future transactions is not included in calculation of EPE;

- (24) ‘counterparty’ for the purposes of Section 7 means any legal or natural person that enters into a netting agreement, and has the contractual capacity to do so;
- (25) ‘contractual cross product netting agreement’ means a bilateral contractual agreement between an institution and a counterparty which creates a single legal obligation (based on netting of covered transactions) covering all bilateral master agreements and transactions belonging to different product categories that are included within the agreement;

For the purposes of this definition, ‘different product categories’ means:

- (a) repurchase transactions, securities and commodities lending and borrowing transactions;
- (b) margin lending transactions;
- (c) the contracts listed in Annex II;

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- (26) ‘payment leg’ means the payment agreed in an OTC derivative transaction with a linear risk profile which stipulates the exchange of a financial instrument for a payment.

In the case of transactions that stipulate the exchange of payment against payment, those two payment legs shall consist of the contractually agreed gross payments, including the notional amount of the transaction.

Section 2**Methods for calculating the exposure value***Article 273***Methods for calculating the exposure value****▼ M8**

1. ► **M17** Institutions shall calculate the exposure value for the contracts listed in Annex II and for credit derivatives, with the exception of the credit derivatives referred to in paragraphs 3 and 5 of this Article, on the basis of one of the methods set out in Sections 3 to 6 in accordance with this Article. ◀

An institution which does not meet the conditions set out in Article 273a(1) shall not use the method set out in Section 4. An institution which does not meet the conditions set out in Article 273a(2) shall not use the method set out in Section 5.

Institutions may use in combination the methods set out in Sections 3 to 6 on a permanent basis within a group. A single institution shall not use in combination the methods set out in Sections 3 to 6 on a permanent basis.

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2. Where permitted by the competent authorities in accordance with Article 283(1) and (2), an institution may determine the exposure value for the following items using the Internal Model Method set out in Section 6:

- (a) the contracts listed in Annex II;
- (b) repurchase transactions;
- (c) securities or commodities lending or borrowing transactions;
- (d) margin lending transactions;
- (e) long settlement transactions.

3. When an institution purchases protection through a credit derivative against a non-trading book exposure or against a counterparty risk exposure, it may calculate its own funds requirement for the hedged exposure in accordance with either of the following:

- (a) Articles 233 to 236;

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- (b) in accordance with Article 183, where permission has been granted in accordance with Article 143.

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The exposure value for CCR for those credit derivatives shall be zero, unless an institution applies the approach in point (h)(ii) of Article 299 (2).

4. Notwithstanding paragraph 3, an institution may choose consistently to include for the purposes of calculating own funds requirements for counterparty credit risk all credit derivatives not included in the trading book and purchased as protection against a non-trading book exposure or against a counterparty credit risk exposure where the credit protection is recognised under this Regulation.

5. Where credit default swaps sold by an institution are treated by an institution as credit protection provided by that institution and are subject to own funds requirement for credit risk of the underlying for the full notional amount, their exposure value for the purposes of CCR in the non-trading book shall be zero.

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- 6. Under the methods set out in Sections 3 to 6, the exposure value for a given counterparty shall be equal to the sum of the exposure values calculated for each netting set with that counterparty.

By way of derogation from the first subparagraph, where one margin agreement applies to multiple netting sets with that counterparty and the institution is using one of the methods set out in Sections 3 to 6 to calculate the exposure value of those netting sets, the exposure value shall be calculated in accordance with the relevant Section.

For a given counterparty, the exposure value for a given netting set of OTC derivative instruments listed in Annex II calculated in accordance with this Chapter shall be the greater of zero and the difference between the sum of exposure values across all netting sets with the counterparty and the sum of credit valuation adjustments for that counterparty being recognised by the institution as an incurred write-down. The credit valuation adjustments shall be calculated without taking into account any offsetting debit value adjustment attributed to the own credit risk of the firm that has been already excluded from own funds in accordance with point (c) of Article 33(1).

- 7. In calculating the exposure value in accordance with the methods set out in Sections 3, 4 and 5, institutions may treat two OTC derivative contracts included in the same netting agreement that are perfectly matching as if they were a single contract with a notional principal equal to zero.

For the purposes of the first subparagraph, two OTC derivative contracts are perfectly matching when they meet all the following conditions:

- (a) their risk positions are opposite;

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- (b) their features, with the exception of the trade date, are identical;
- (c) their cash flows fully offset each other.

8. Institutions shall determine the exposure value for exposures arising from long settlement transactions by any of the methods set out in Sections 3 to 6 of this Chapter, regardless of which method the institution has chosen for treating OTC derivatives and repurchase transactions, securities or commodities lending or borrowing transactions, and margin lending transactions. In calculating the own funds requirements for long settlement transactions, an institution that uses the approach set out in Chapter 3 may assign the risk weights under the approach set out in Chapter 2 on a permanent basis and irrespective of the materiality of those positions.

9. For the methods set out in Sections 3 to 6 of this Chapter, institutions shall treat transactions where Specific Wrong-Way risk has been identified in accordance with Article 291(2), (4), (5), and (6).

*Article 273a***Conditions for using simplified methods for calculating the exposure value**

1. An institution may calculate the exposure value of its derivative positions in accordance with the method set out in Section 4, provided that the size of its on- and off-balance-sheet derivative business is equal to or less than both of the following thresholds on the basis of an assessment carried out on a monthly basis using the data as of the last day of the month:

- (a) 10 % of the institution's total assets;
- (b) EUR 300 million.

2. An institution may calculate the exposure value of its derivative positions in accordance with the method set out in Section 5, provided that the size of its on- and off-balance-sheet derivative business is equal to or less than both of the following thresholds on the basis of an assessment carried out on a monthly basis using the data as of the last day of the month:

- (a) 5 % of the institution's total assets;
- (b) EUR 100 million.

3. For the purposes of paragraphs 1 and 2, institutions shall calculate the size of their on- and off-balance-sheet derivative business on the basis of data as of the last day of each month in accordance with the following requirements:

- (a) derivative positions shall be valued at their market values on that given date; where the market value of a position is not available on a given date, institutions shall take a fair value for the position on that date; where the market value and fair value of a position are not available on a given date, institutions shall take the most recent of the market value or fair value for that position;

▼ M17

- (b) the absolute value of the aggregated long position shall be summed with the absolute value of the aggregated short position;

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- (c) all derivative positions shall be included, except credit derivatives that are recognised as internal hedges against non-trading book credit risk exposures.

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For the purposes of the first subparagraph, the meaning of long and short positions is the same as that set out in Article 94(3).

For the purposes of the first subparagraph, the value of the aggregated long (short) position shall be equal to the sum of the values of the individual long (short) positions included in the calculation in accordance with point (c).

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4. By way of derogation from paragraph 1 or 2, as applicable, where the derivative business on a consolidated basis does not exceed the thresholds set out in paragraph 1 or 2, as applicable, an institution which is included in the consolidation and which would have to apply the method set out in Section 3 or 4 because it exceeds those thresholds on an individual basis, may, subject to the approval of competent authorities, instead choose to apply the method that would apply on a consolidated basis.

5. Institutions shall notify the competent authorities of the methods set out in Section 4 or 5 that they use, or cease to use, as applicable, to calculate the exposure value of their derivative positions.

6. Institutions shall not enter into a derivative transaction or buy or sell a derivative instrument for the sole purpose of complying with any of the conditions set out in paragraphs 1 and 2 during the monthly assessment.

*Article 273b***▼ M17**

Non-compliance with the conditions for using simplified methods for calculating the exposure value of derivatives and the simplified approach for calculating the own funds requirements for CVA risk

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1. An institution that no longer meets one or more of the conditions set out in Article 273a(1) or (2) shall immediately notify the competent authority thereof.

2. ► **M17** Institutions shall cease to calculate the exposure values of their derivative positions in accordance with Section 4 or 5 and to calculate the own funds requirements for CVA risk in accordance with Article 385, as applicable, within three months of the occurrence of one of the following: ◀

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- (a) the institution does not meet the conditions set out in point (a) of Article 273a(1) or (2), as applicable, or the conditions set out in point (b) of Article 273a(1) or (2), as applicable, for three consecutive months;
- (b) the institution does not meet the conditions set out in point (a) of Article 273a(1) or (2), as applicable, or the conditions set out in point (b) of Article 273a(1) or (2), as applicable, for more than six of the preceding 12 months.

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3. Institutions that have ceased to calculate the exposure values of their derivative positions in accordance with Section 4 or 5 and to calculate the own funds requirements for CVA risk in accordance with Article 385, as applicable, shall only be permitted to resume calculating the exposure value of their derivative positions as set out in Section 4 or 5 and the own funds requirements for CVA risk in accordance with Article 385 where they demonstrate to the competent authority that all of the conditions set out in Article 273a(1) or (2), have been met for an uninterrupted period of one year.

▼ M8**Section 3****Standardised approach for counterparty credit risk***Article 274***Exposure value**

1. An institution may calculate a single exposure value at netting set level for all the transactions covered by a contractual netting agreement where all the following conditions are met:

- (a) the netting agreement belongs to one of the types of contractual netting agreements referred to in Article 295;
- (b) the netting agreement has been recognised by competent authorities in accordance with Article 296;
- (c) the institution has fulfilled the obligations laid down in Article 297 in respect of the netting agreement.

Where any of the conditions set out in the first subparagraph are not met, the institution shall treat each transaction as if it was its own netting set.

2. Institutions shall calculate the exposure value of a netting set under the standardised approach for counterparty credit risk as follows:

$$\text{Exposure value} = \alpha \cdot (\text{RC} + \text{PFE})$$

where:

RC = the replacement cost calculated in accordance with Article 275;
and

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PFE = the potential future exposure calculated in accordance with Article 278;

α = 1,4.

3. The exposure value of a netting set that is subject to a contractual margin agreement shall be capped at the exposure value of the same netting set not subject to any form of margin agreement.

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4. Where multiple margin agreements apply to the same netting set, or the same netting set includes both transactions subject to a margin agreement and transactions not subject to a margin agreement, an institution shall calculate its exposure value as follows:

(a) the institution shall establish the hypothetical sub-netting sets concerned, composed of transactions included in the netting set, as follows:

(i) all transactions subject to a margin agreement and to the same margin period of risk as determined in accordance with Article 285(2) to (5), shall be allocated to the same sub-netting set;

(ii) all transactions not subject to a margin agreement shall be allocated to the same sub-netting set, distinct from the sub-netting sets established in accordance with point (i) of this paragraph;

(b) the institution shall calculate the replacement cost of the netting set in accordance with Article 275(2), taking into account all transactions within the netting set, whether or not subject to a margin agreement, and apply all of the following:

(i) CMV shall be calculated for all transactions within a netting set gross of any collateral held or posted where positive and negative market values are netted in computing the CMV;

(ii) NICA, VM, TH, and MTA, where applicable, shall be calculated separately as the sum across the same inputs applicable to each individual margin agreement of the netting set;

(c) the institution shall calculate the potential future exposure of the netting set referred to in Article 278 by applying all of the following:

(i) the multiplier referred to in Article 278(1) shall be based on the inputs CMV, NICA and VM, as applicable, in accordance with point (b) of this paragraph;

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- (ii) $\sum_a AddOn^{(a)}$ shall be calculated in accordance with Article 278, separately for each hypothetical sub-netting set referred to in point (a) of this paragraph.

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5. Institutions may set to zero the exposure value of a netting set that satisfies all the following conditions:

- (a) the netting set is solely composed of sold options;
- (b) the current market value of the netting set is at all times negative;
- (c) the premium of all the options included in the netting set has been received upfront by the institution to guarantee the performance of the contracts;
- (d) the netting set is not subject to any margin agreement.

6. In a netting set, institutions shall replace a transaction which is a finite linear combination of bought or sold call or put options with all the single options that form that linear combination, taken as an individual transaction, for the purpose of calculating the exposure value of the netting set in accordance with this Section. Each such combination of options shall be treated as an individual transaction in the netting set in which the combination is included for the purpose of calculating the exposure value.

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By way of derogation from the first subparagraph, institutions shall replace a vanilla digital option the strike of which equals K with the relevant collar combination of two sold and bought vanilla call or put options that meet the following requirements:

- (a) the two options of the collar combination have:
 - (i) the same expiry date and the same spot or forward price of the underlying instrument as the vanilla digital option;
 - (ii) strikes equal to $0,95 \cdot K$ and $1,05 \cdot K$ respectively;
- (b) the collar combination replicates exactly the vanilla digital option payoff outside the range between the two strikes referred to in point (a).

The risk position of the two options of the collar combination referred to in the second subparagraph shall be calculated separately in accordance with Article 279.

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7. The exposure value of a credit derivative transaction representing a long position in the underlying may be capped to the amount of outstanding unpaid premium provided it is treated as its own netting set that is not subject to a margin agreement.

▼ **M8***Article 275***Replacement cost**

1. Institutions shall calculate the replacement cost RC for netting sets that are not subject to a margin agreement, in accordance with the following formula:

$$RC = \max\{CMV - NICA, 0\}$$

2. Institutions shall calculate the replacement cost for single netting sets that are subject to a margin agreement in accordance with the following formula:

$$RC = \max\{CMV - VM - NICA, TH + MTA - NICA, 0\}$$

where:

RC = the replacement cost;

VM = the volatility-adjusted value of the net variation margin received or posted, as applicable, to the netting set on a regular basis to mitigate changes in the netting set's CMV;

TH = the margin threshold applicable to the netting set under the margin agreement below which the institution cannot call for collateral; and

MTA = the minimum transfer amount applicable to the netting set under the margin agreement.

3. Institutions shall calculate the replacement cost for multiple netting sets that are subject to the same margin agreement in accordance with the following formula:

$$RC = \max\left\{\sum_i \max\{CMV_i, 0\} - \max\{VM_{MA} + NICA_{MA}, 0\}, 0\right\} + \max\left\{\sum_i \min\{CMV_i, 0\} - \min\{VM_{MA} + NICA_{MA}, 0\}, 0\right\}$$

where:

RC = the replacement cost;

i = the index that denotes the netting sets that are subject to the single margin agreement;

CMV_i = the CMV of netting set i;

VM_{MA} = the sum of the volatility-adjusted value of collateral received or posted, as applicable, to multiple netting sets on a regular basis to mitigate changes in their CMV; and

NICA_{MA} = the sum of the volatility-adjusted value of collateral received or posted, as applicable, to multiple netting sets other than VM_{MA}.

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For the purposes of the first subparagraph, $NICA_{MA}$ may be calculated at trade level, at netting set level or at the level of all the netting sets to which the margin agreement applies depending on the level at which the margin agreement applies.

*Article 276***Recognition and treatment of collateral**

1. For the purposes of this Section, institutions shall calculate the collateral amounts of VM, VM_{MA} , NICA and $NICA_{MA}$, by applying all the following requirements:

- (a) where all the transactions included in a netting set belong to the trading book, only collateral that is eligible under Articles 197 and 299 shall be recognised;
- (b) where a netting set contains at least one transaction that belongs to the non-trading book, only collateral that is eligible under Article 197 shall be recognised;
- (c) collateral received from a counterparty shall be recognised with a positive sign and collateral posted to a counterparty shall be recognised with a negative sign;

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- (d) the volatility-adjusted value of any type of collateral received or posted shall be calculated in accordance with Article 223;

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- (e) the same collateral item shall not be included in both VM and NICA at the same time;
- (f) the same collateral item shall not be included in both VM_{MA} and $NICA_{MA}$ at the same time;
- (g) any collateral posted to the counterparty that is segregated from the assets of that counterparty and, as a result of that segregation, is bankruptcy remote in the event of the default or insolvency of that counterparty shall not be recognised in the calculation of NICA and $NICA_{MA}$.

2. For the calculation of the volatility-adjusted value of collateral posted referred to in point (d) of paragraph 1 of this Article, institutions shall replace the formula set out in Article 223(2) with the following formula:

$$C_{VA} = C \cdot (1 + H_C + H_{fx})$$

where:

C_{VA} = the volatility-adjusted value of collateral posted; and

C = the collateral;

H_C and H_{fx} are defined in accordance with Article 223(2).

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3. For the purposes of point (d) of paragraph 1, institutions shall set the liquidation period relevant for the calculation of the volatility-adjusted value of any collateral received or posted in accordance with one of the following time horizons:

- (a) one year for the netting sets referred to in Article 275(1);
- (b) the margin period of risk determined in accordance with point (b) of Article 279c(1) for the netting sets referred to in Article 275(2) and (3).

*Article 277***Mapping of transactions to risk categories**

1. Institutions shall map each transaction of a netting set to one of the following risk categories to determine the potential future exposure of the netting set referred to in Article 278:

- (a) interest rate risk;
- (b) foreign exchange risk;
- (c) credit risk;
- (d) equity risk;
- (e) commodity risk;
- (f) other risks.

2. Institutions shall conduct the mapping referred to in paragraph 1 on the basis of the primary risk driver of a derivative transaction. The primary risk driver shall be the only material risk driver of a derivative transaction.

3. By way of derogation from paragraph 2, institutions shall map derivative transactions that have more than one material risk driver to more than one risk category. Where all the material risk drivers of one of those transactions belong to the same risk category, institutions shall only be required to map that transaction once to that risk category on the basis of the most material of those risk drivers. Where the material risk drivers of one of those transactions belong to different risk categories, institutions shall map that transaction once to each risk category for which the transaction has at least one material risk driver, on the basis of the most material of the risk drivers in that risk category.

4. Notwithstanding paragraphs 1, 2 and 3, when mapping transactions to the risk categories listed in paragraph 1, institutions shall apply the following requirements:

- (a) where the primary risk driver of a transaction, or the most material risk driver in a given risk category for transactions referred to in paragraph 3, is an inflation variable, institutions shall map the transaction to the interest rate risk category;

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- (b) where the primary risk driver of a transaction, or the most material risk driver in a given risk category for transactions referred to in paragraph 3, is a climatic conditions variable, institutions shall map the transaction to the commodity risk category.

5. EBA shall develop draft regulatory technical standards to specify:

- (a) the method for identifying transactions with only one material risk driver;
- (b) the method for identifying transactions with more than one material risk driver and for identifying the most material of those risk drivers for the purposes of paragraph 3.

EBA shall submit those draft regulatory technical standards to the Commission by 28 December 2019.

Power is delegated to the Commission to supplement this Regulation by adopting the regulatory technical standards referred to in the first subparagraph in accordance with Articles 10 to 14 of Regulation (EU) No 1093/2010.

Article 277a

Hedging sets

1. Institutions shall establish the relevant hedging sets for each risk category of a netting set and assign each transaction to those hedging sets as follows:

- (a) transactions mapped to the interest rate risk category shall be assigned to the same hedging set only where their primary risk driver, or the most material risk driver in the given risk category for transactions referred to in Article 277(3), is denominated in the same currency;
- (b) transactions mapped to the foreign exchange risk category shall be assigned to the same hedging set only where their primary risk driver, or the most material risk driver in the given risk category for transactions referred to in Article 277(3), is based on the same currency pair;
- (c) all the transactions mapped to the credit risk category shall be assigned to the same hedging set;
- (d) all the transactions mapped to the equity risk category shall be assigned to the same hedging set;
- (e) transactions mapped to the commodity risk category shall be assigned to one of the following hedging sets on the basis of the nature of their primary risk driver or the most material risk driver in the given risk category for transactions referred to in Article 277(3):
 - (i) energy;
 - (ii) metals;
 - (iii) agricultural goods;
 - (iv) other commodities;
 - (v) climatic conditions;

▼ **M8**

- (f) transactions mapped to the other risks category shall be assigned to the same hedging set only where their primary risk driver, or the most material risk driver in the given risk category for transactions referred to in Article 277(3), is identical.

For the purposes of point (a) of the first subparagraph of this paragraph, transactions mapped to the interest rate risk category that have an inflation variable as the primary risk driver shall be assigned to separate hedging sets, other than the hedging sets established for transactions mapped to the interest rate risk category that do not have an inflation variable as the primary risk driver. Those transactions shall be assigned to the same hedging set only where their primary risk driver, or the most material risk driver in the given risk category for transactions referred to in Article 277(3), is denominated in the same currency.

2. By way of derogation from paragraph 1 of this Article, institutions shall establish separate individual hedging sets in each risk category for the following transactions:

- (a) transactions for which the primary risk driver, or the most material risk driver in the given risk category for transactions referred to in Article 277(3), is either the market implied volatility or the realised volatility of a risk driver or the correlation between two risk drivers;
- (b) transactions for which the primary risk driver, or the most material risk driver in the given risk category for transactions referred to in Article 277(3), is the difference between two risk drivers mapped to the same risk category or transactions that consist of two payment legs denominated in the same currency and for which a risk driver from the same risk category of the primary risk driver is contained in the other payment leg than the one containing the primary risk driver.

For the purposes of point (a) of the first subparagraph of this paragraph, institutions shall assign transactions to the same hedging set of the relevant risk category only where their primary risk driver, or the most material risk driver in the given risk category for transactions referred to in Article 277(3), is identical.

For the purposes of point (b) of the first subparagraph, institutions shall assign transactions to the same hedging set of the relevant risk category only where the pair of risk drivers in those transactions as referred to therein is identical and the two risk drivers contained in this pair are positively correlated. Otherwise, institutions shall assign transactions referred to in point (b) of the first subparagraph to one of the hedging sets established in accordance with paragraph 1, on the basis of only one of the two risk drivers referred to in point (b) of the first subparagraph.

▼ M17

For the purposes of the first subparagraph, point (a), of this paragraph, institutions shall assign transactions to a separate hedging set of the relevant risk category following the same hedging set construction set out in paragraph 1.

▼ M8

3. Institutions shall make available upon request by the competent authorities the number of hedging sets established in accordance with paragraph 2 of this Article for each risk category, with the primary risk driver, or the most material risk driver in the given risk category for transactions referred to in Article 277(3), or the pair of risk drivers of each of those hedging sets and with the number of transactions in each of those hedging sets.

*Article 278***Potential future exposure**

1. Institutions shall calculate the potential future exposure of a netting set as follows:

$$\text{PFE} = \text{multiplier} \cdot \sum_a \text{AddOn}^{(a)}$$

where:

PFE = the potential future exposure;

a = the index that denotes the risk categories included in the calculation of the potential future exposure of the netting set;

AddOn^(a) = the add-on for risk category a calculated in accordance with Articles 280a to 280f, as applicable; and

multiplier = the multiplication factor calculated in accordance with the formula referred to in paragraph 3.

For the purpose of this calculation, institutions shall include the add-on of a given risk category in the calculation of the potential future exposure of a netting set where at least one transaction of the netting set has been mapped to that risk category.

2. The potential future exposure of multiple netting sets that are subject to one margin agreement, as referred in Article 275(3), shall be calculated as the sum of the potential future exposures of all the individual netting sets as if they were not subject to any form of a margin agreement.

3. For the purposes of paragraph 1, the multiplier shall be calculated as follows:

$$\text{multiplier} = \begin{cases} 1 & \text{if } z \geq 0 \\ \min\{1, \text{Floor}_m + (1 - \text{Floor}_m) \cdot \exp\left(\frac{z}{y}\right)\} & \text{if } z < 0 \end{cases}$$

where:

Floor_m = 5 %;

▼M8

$$y = 2 \cdot (1 - \text{Floor}_m) \cdot \Sigma_a \text{AddOn}^{(a)}$$

$$z = \begin{cases} \text{CMV} - \text{NICA} & \text{for the netting sets referred to in Article 275(1)} \\ \text{CMV} - \text{VM} - \text{NICA} & \text{for the netting sets referred to in Article 275(2)} \\ \text{CMV}_i - \text{NICA}_i & \text{for the netting sets referred to in Article 275(3)} \end{cases}$$

NICA_i = the net independent collateral amount calculated only for transactions that are included in netting set i . NICA_i shall be calculated at trade level or at netting set level depending on the margin agreement.

*Article 279***Calculation of the risk position**

For the purpose of calculating the risk category add-ons referred to in Articles 280a to 280f, institutions shall calculate the risk position of each transaction of a netting set as follows:

$$\text{RiskPosition} = \delta \cdot \text{AdjNot} \cdot \text{MF}$$

where:

δ = the supervisory delta of the transaction calculated in accordance with the formula laid down in Article 279a;

AdjNot = the adjusted notional amount of the transaction calculated in accordance with Article 279b; and

MF = the maturity factor of the transaction calculated in accordance with the formula laid down in Article 279c.

*Article 279a***Supervisory delta**

1. Institutions shall calculate the supervisory delta as follows:

▼M17

- (a) for call and put options that entitle the option buyer to purchase or sell an underlying instrument at a positive price on a single or multiple dates in the future, except where those options are mapped to the interest rate risk or commodity risk category, institutions shall use the following formula:

▼M8

$$\delta = \text{sign} \cdot N \left(\text{type} \cdot \frac{\ln(P/K) + 0,5 \cdot \sigma^2 \cdot T}{\sigma \cdot \sqrt{T}} \right)$$

where:

δ = the supervisory delta;

sign = -1 where the transaction is a sold call option or a bought put option;

▼ M8

sign = + 1 where the transaction is a bought call option or sold put option;

type = – 1 where the transaction is a put option;

type = + 1 where the transaction is a call option;

$N(x)$ = the cumulative distribution function for a standard normal random variable meaning the probability that a normal random variable with mean zero and variance of one is less than or equal to x ;

P = the spot or forward price of the underlying instrument of the option; for options the cash flows of which depend on an average value of the price of the underlying instrument, P shall be equal to the average value at the calculation date;

K = the strike price of the option;

▼ C7

T = the period between the expiry date of the option (T_{exp}) and the reporting date; for options which can be exercised at one future date only, T_{exp} is equal to that date; for options which can be exercised at multiple future dates, T_{exp} is equal to the latest of those dates; T shall be expressed in years using the relevant business day convention; and

▼ M8

σ = the supervisory volatility of the option determined in accordance with Table 1 on the basis of the risk category of the transaction and the nature of the underlying instrument of the option.

Table 1

Risk category	Underlying instrument	Supervisory volatility
Foreign exchange	All	15 %
Credit	Single-name instrument	100 %
	Multiple-names instrument	80 %
Equity	Single-name instrument	120 %
	Multiple-names instrument	75 %
Commodity	Electricity	150 %
	Other commodities (excluding electricity)	70 %
Others	All	150 %

Institutions using the forward price of the underlying instrument of an option shall ensure that:

- (i) the forward price is consistent with the characteristics of the option;

▼M8

- (ii) the forward price is calculated using a relevant interest rate prevailing at the reporting date;
 - (iii) the forward price integrates the expected cash flows of the underlying instrument before the expiry of the option;
- (b) for tranches of a synthetic securitisation and a nth-to-default credit derivative, institutions shall use the following formula:

$$\delta = \text{sign} \cdot \frac{15}{(1 + 14 \cdot A) \cdot (1 + 14 \cdot D)}$$

where:

$$\text{sign} = \begin{cases} +1 & \text{where credit protection has been obtained through the transaction} \\ -1 & \text{where credit protection has been provided through the transaction} \end{cases}$$

A = the attachment point of the tranche; for a nth-to-default credit derivative transaction based on reference entities k, $A = (n - 1)/k$; and

D = the detachment point of the tranche; for a nth-to-default credit derivative transaction based on reference entities k, $D = n/k$;

- (c) for transactions not referred to in point (a) or (b), institutions shall use the following supervisory delta:

$$\delta = \begin{cases} +1 & \text{if the transaction is a long position in the primary risk driver or in the most material risk driver in the given risk category} \\ -1 & \text{if the transaction is a short position in the primary risk driver or in the most material risk driver in the given risk category} \end{cases}$$

2. For the purposes of this Section, a long position in the primary risk driver or in the most material risk driver in the given risk category for transactions referred to in Article 277(3) means that the market value of the transaction increases when the value of that risk driver increases and a short position in the primary risk driver or in the most material risk driver in the given risk category for transactions referred to in Article 277(3) means that the market value of the transaction decreases when the value of that risk driver increases.

3. EBA shall develop draft regulatory technical standards to specify:

▼M17

- (a) in accordance with international regulatory developments, the formulae that institutions shall use to calculate the supervisory delta of call and put options mapped to the interest rate risk or commodity risk category compatible with market conditions in which interest rates or commodity prices may be negative and the supervisory volatility that is suitable for those formulae;

▼ M8

- (b) the method for determining whether a transaction is a long or short position in the primary risk driver or in the most material risk driver in the given risk category for transactions referred to in Article 277(3).

▼ M17

EBA shall submit those draft regulatory technical standards to the Commission by 10 July 2025.

▼ M8

Power is delegated to the Commission to supplement this Regulation by adopting the regulatory technical standards referred to in the first subparagraph in accordance with Articles 10 to 14 of Regulation (EU) No 1093/2010.

*Article 279b***Adjusted notional amount**

1. Institutions shall calculate the adjusted notional amount as follows:

- (a) for transactions mapped to the interest rate risk category or the credit risk category, institutions shall calculate the adjusted notional amount as the product of the notional amount of the derivative contract and the supervisory duration factor, which shall be calculated as follows:

▼ C7

$$\text{supervisory duration factor} = \max \left\{ \frac{\exp(-R \cdot S) - \exp(-R \cdot E)}{R}; 10/\text{OneBusinessYear} \right\}$$

where:

- R = the supervisory discount rate; R = 5 %;
- S = the period between the start date of a transaction and the reporting date, which shall be expressed in years using the relevant business day convention;
- E = the period between the end date of a transaction and the reporting date, which shall be expressed in years using the relevant business day convention; and

OneBusinessYear = one year expressed in business days using the relevant business day convention.

▼ M8

The start date of a transaction is the earliest date at which at least a contractual payment under the transaction, to or from the institution, is either fixed or exchanged, other than payments related to the exchange of collateral in a margin agreement. Where the transaction has already been fixing or making payments at the reporting date, the start date of a transaction shall be equal to 0.

Where a transaction involves one or more contractual future dates on which the institution or the counterparty may decide to terminate the transaction prior to its contractual maturity, the start date of a transaction shall be equal to the earliest of the following:

- (i) the date or the earliest of the multiple future dates at which the institution or the counterparty may decide to terminate the transaction earlier than its contractual maturity;

▼M8

- (ii) the date at which a transaction starts fixing or making payments, other than payments related to the exchange of collateral in a margin agreement.

Where a transaction has a financial instrument as the underlying instrument that may give rise to contractual obligations additional to those of the transaction, the start date of a transaction shall be determined on the basis of the earliest date at which the underlying instrument starts fixing or making payments.

The end date of a transaction is the latest date at which a contractual payment under the transaction, to or from the institution, is or may be exchanged.

Where a transaction has a financial instrument as an underlying instrument that may give rise to contractual obligations additional to those of the transaction, the end date of a transaction shall be determined on the basis of the last contractual payment of the underlying instrument of the transaction.

Where a transaction is structured to settle an outstanding exposure following specified payment dates and where the terms are reset so that the market value of the transaction is zero on those specified dates, the settlement of the outstanding exposure at those specified dates is considered a contractual payment under the same transaction;

- (b) for transactions mapped to the foreign exchange risk category, institutions shall calculate the adjusted notional amount as follows:
 - (i) where the transaction consists of one payment leg, the adjusted notional amount shall be the notional amount of the derivative contract;
 - (ii) where the transaction consists of two payment legs and the notional amount of one payment leg is denominated in the institution's reporting currency, the adjusted notional amount shall be the notional amount of the other payment leg;
 - (iii) where the transaction consists of two payment legs and the notional amount of each payment leg is denominated in a currency other than the institution's reporting currency, the adjusted notional amount shall be the largest of the notional amounts of the two payment legs after those amounts have been converted into the institution's reporting currency at the prevailing spot exchange rate;
- (c) for transactions mapped to the equity risk category or commodity risk category, institutions shall calculate the adjusted notional amount as the product of the market price of one unit of the underlying instrument of the transaction and the number of units in the underlying instrument referenced by the transaction;

▼M8

where a transaction mapped to the equity risk category or commodity risk category is contractually expressed as a notional amount, institutions shall use the notional amount of the transaction rather than the number of units in the underlying instrument as the adjusted notional amount;

- (d) for transactions mapped to the other risks category, institutions shall calculate the adjusted notional amount on the basis of the most appropriate method among the methods set out in points (a), (b) and (c), depending on the nature and characteristics of the underlying instrument of the transaction.

2. Institutions shall determine the notional amount or number of units of the underlying instrument for the purpose of calculating the adjusted notional amount of a transaction referred to in paragraph 1 as follows:

- (a) where the notional amount or the number of units of the underlying instrument of a transaction is not fixed until its contractual maturity:
 - (i) for deterministic notional amounts and numbers of units of the underlying instrument, the notional amount shall be the weighted average of all the deterministic values of notional amounts or number of units of the underlying instrument, as applicable, until the contractual maturity of the transaction, where the weights are the proportion of the time period during which each value of notional amount applies;
 - (ii) for stochastic notional amounts and numbers of units of the underlying instrument, the notional amount shall be the amount determined by fixing current market values within the formula for calculating the future market values;
- (b) for contracts with multiple exchanges of the notional amount, the notional amount shall be multiplied by the number of remaining payments still to be made in accordance with the contracts;
- (c) for contracts that provide for a multiplication of the cash-flow payments or a multiplication of the underlying of the derivative contract, the notional amount shall be adjusted by an institution to take into account the effects of the multiplication on the risk structure of those contracts.

3. Institutions shall convert the adjusted notional amount of a transaction into their reporting currency at the prevailing spot exchange rate where the adjusted notional amount is calculated under this Article from a contractual notional amount or a market price of the number of units of the underlying instrument denominated in another currency.

▼M8*Article 279c***Maturity Factor**

1. Institutions shall calculate the maturity factor as follows:
 - (a) for transactions included in the netting sets referred to in Article 275(1), institutions shall use the following formula:

$$MF = \sqrt{\min\{\max\{M, 10/\text{OneBusinessYear}\}, 1\}}$$

where:

MF = the maturity factor;

M = the remaining maturity of the transaction which is equal to the period of time needed for the termination of all contractual obligations of the transaction; for that purpose, any optionality of a derivative contract shall be considered to be a contractual obligation; the remaining maturity shall be expressed in years using the relevant business day convention;

where a transaction has another derivative contract as underlying instrument that may give rise to additional contractual obligations beyond the contractual obligations of the transaction, the remaining maturity of the transaction shall be equal to the period of time needed for the termination of all contractual obligations of the underlying instrument;

where a transaction is structured to settle outstanding exposure following specified payment dates and where the terms are reset so that the market value of the transaction is zero on those specified dates, the remaining maturity of the transaction shall be equal to the time until the next reset date; and

OneBusinessYear = one year expressed in business days using the relevant business day convention;

- (b) for transactions included in the netting sets referred to in Article 275(2) and (3), the maturity factor is defined as:

▼ M8

$$MF = \frac{3}{2} \sqrt{\frac{MPOR}{OneBusinessYear}}$$

where:

MF = the maturity factor;

MPOR = the margin period of risk of the netting set determined in accordance with Article 285 (2) to (5); and

OneBusinessYear = one year expressed in business days using the relevant business day convention.

When determining the margin period of risk for transactions between a client and a clearing member, an institution acting either as the client or as the clearing member shall replace the minimum period set out in point (b) of Article 285(2) with five business days.

2. For the purposes of paragraph 1, the remaining maturity shall be equal to the period of time until the next reset date for transactions that are structured to settle outstanding exposure following specified payment dates and where the terms are reset in such a way that the market value of the contract shall be zero on those specified payment dates.

Article 280

Hedging set supervisory factor coefficient

For the purpose of calculating the add-on of a hedging set as referred to in Articles 280a to 280f, the hedging set supervisory factor coefficient 'e' shall be the following:

$$e = \begin{cases} 1 & \text{for the hedging sets established in accordance with Article 277a(1)} \\ 5 & \text{for the hedging sets established in accordance with point (a) of Article 277a(2)} \\ 0,5 & \text{for the hedging sets established in accordance with point (b) of Article 277a(2)} \end{cases}$$

Article 280a

Interest rate risk category add-on

1. For the purposes of Article 278, institutions shall calculate the interest rate risk category add-on for a given netting set as follows:

$$AddOn^{IR} = \sum_j AddOn_j^{IR}$$

where:

$AddOn^{IR}$ = the interest rate risk category add-on;

▼ M8

j = the index that denotes all the interest rate risk hedging sets established in accordance with point (a) of Article 277a(1) and with Article 277a(2) for the netting set; and

$\text{AddOn}_j^{\text{IR}}$ = the interest rate risk category add-on for hedging set j calculated in accordance with paragraph 2.

2. Institutions shall calculate the interest rate risk category add-on for hedging set j as follows:

$$\text{AddOn}_j^{\text{IR}} = \epsilon_j \cdot \text{SF}^{\text{IR}} \cdot \text{EffNot}_j^{\text{IR}}$$

where:

ϵ_j = the hedging set supervisory factor coefficient of hedging set j determined in accordance with the applicable value specified in Article 280;

SF^{IR} = the supervisory factor for the interest rate risk category with a value equal to 0,5 %; and

$\text{EffNot}_j^{\text{IR}}$ = the effective notional amount of hedging set j calculated in accordance with paragraph 3.

3. For the purpose of calculating the effective notional amount of hedging set j , institutions shall first map each transaction of the hedging set to the appropriate bucket in Table 2. They shall do so on the basis of the end date of each transaction as determined under point (a) of Article 279b(1):

Table 2

Bucket	End date (in years)
1	> 0 and ≤ 1
2	> 1 and ≤ 5
3	> 5

Institutions shall then calculate the effective notional amount of hedging set j in accordance with the following formula:

▼ C7

$$\text{EffNot}_j^{\text{IR}} = \sqrt{[(D_{j,1})^2 + (D_{j,2})^2 + (D_{j,3})^2 + 1,4 \cdot D_{j,1} \cdot D_{j,2} + 1,4 \cdot D_{j,2} \cdot D_{j,3} + 0,6 \cdot D_{j,1} \cdot D_{j,3}]}$$

▼ M8

where:

$\text{EffNot}_j^{\text{IR}}$ = the effective notional amount of hedging set j ; and

$D_{j,k}$ = the effective notional amount of bucket k of hedging set j calculated as follows:

$$D_{j,k} = \sum_{l \in \text{Bucket } k} \text{RiskPosition}_l$$

where:

l = the index that denotes the risk position.

▼M8*Article 280b***Foreign exchange risk category add-on**

1. For the purposes of Article 278, institutions shall calculate the foreign exchange risk category add-on for a given netting set as follows:

$$\text{AddOn}^{\text{FX}} = \sum_j \text{AddOn}_j^{\text{FX}}$$

where:

AddOn^{FX} = the foreign exchange risk category add on;

j = the index that denotes the foreign exchange risk hedging sets established in accordance with point (b) of Article 277a(1) and with Article 277a (2) for the netting set; and

$\text{AddOn}_j^{\text{FX}}$ = the foreign exchange risk category add-on for hedging set j calculated in accordance with paragraph 2.

2. Institutions shall calculate the foreign exchange risk category add-on for hedging set j as follows:

$$\text{AddOn}_j^{\text{FX}} = e_j \cdot \text{SF}^{\text{FX}} \cdot |\text{EffNot}_j^{\text{FX}}|$$

where:

e_j = the hedging set supervisory factor coefficient of hedging set j determined in accordance with Article 280;

SF^{FX} = the supervisory factor for the foreign exchange risk category with a value equal to 4 %;

$\text{EffNot}_j^{\text{FX}}$ = the effective notional amount of hedging set j calculated as follows:

$$\text{EffNot}_j^{\text{FX}} = \sum_{l \in \text{Hedging set } j} \text{RiskPosition}_l$$

where:

l = the index that denotes the risk position.

*Article 280c***Credit risk category add-on**

1. For the purposes of paragraph 2, institutions shall establish the relevant credit reference entities of the netting set in accordance with the following:

▼M8

- (a) there shall be one credit reference entity for each issuer of a reference debt instrument that underlies a single-name transaction allocated to the credit risk category; single-name transactions shall be assigned to the same credit reference entity only where the underlying reference debt instrument of those transactions is issued by the same issuer;
- (b) there shall be one credit reference entity for each group of reference debt instruments or single-name credit derivatives that underlie a multi-name transaction allocated to the credit risk category; multi-names transactions shall be assigned to the same credit reference entity only where the group of underlying reference debt instruments or single-name credit derivatives of those transactions have the same constituents.

2. For the purposes of Article 278, institution shall calculate the credit risk category add-on for a given netting set as follows:

$$\text{AddOn}^{\text{Credit}} = \sum_j \text{AddOn}_j^{\text{Credit}}$$

where:

$\text{AddOn}^{\text{Credit}}$ = credit risk category add-on;

j = the index that denotes all the credit risk hedging sets established in accordance with point (c) of Article 277a(1) and with Article 277a(2) for the netting set; and

$\text{AddOn}_j^{\text{Credit}}$ = the credit risk category add-on for hedging set j calculated in accordance with paragraph 3.

3. Institutions shall calculate the credit risk category add-on for hedging set j as follows:

▼C7

$$\text{AddOn}_j^{\text{Credit}} = \epsilon_j \sqrt{\left(\sum_k \rho_k^{\text{Credit}} \cdot \text{AddOn}(\text{Entity}_k) \right)^2 + \sum_k \left(1 - (\rho_k^{\text{Credit}})^2 \right) \cdot (\text{AddOn}(\text{Entity}_k))^2}$$

▼M8

where:

$\text{AddOn}_j^{\text{Credit}}$ = the credit risk category add-on for hedging set j ;

ϵ_j = the hedging set supervisory factor coefficient of hedging set j determined in accordance with Article 280;

k = the index that denotes the credit reference entities of the netting set established in accordance with paragraph 1;

▼M8

ρ_k^{Credit} = the correlation factor of the credit reference entity k; where the credit reference entity k has been established in accordance with point (a) of paragraph 1, $\rho_k^{\text{Credit}} = 50\%$, where the credit reference entity k has been established in accordance with point (b) of paragraph 1, $\rho_k^{\text{Credit}} = 80\%$; and

$\text{AddOn}(\text{Entity}_k)$ = the add-on for the credit reference entity k determined in accordance with paragraph 4.

4. Institutions shall calculate the add-on for the credit reference entity k as follows:

$$\text{AddOn}(\text{Entity}_k) = \text{EffNot}_k^{\text{Credit}}$$

where:

$\text{EffNot}_k^{\text{Credit}}$ = the effective notional amount of the credit reference entity k calculated as follows:

$$\text{EffNot}_k^{\text{Credit}} = \sum_{l \in \text{Credit reference entity k}} \text{SF}_{k,l}^{\text{Credit}} \cdot \text{RiskPosition}_l$$

where:

l = the index that denotes the risk position; and

$\text{SF}_{k,l}^{\text{Credit}}$ = the supervisory factor applicable to the credit reference entity k calculated in accordance with paragraph 5.

5. Institutions shall calculate the supervisory factor applicable to the credit reference entity k as follows:

- (a) for the credit reference entity k established in accordance with point (a) of paragraph 1, $\text{SF}_{k,l}^{\text{Credit}}$ shall be mapped to one of the six supervisory factors set out in Table 3 of this paragraph on the basis of an external credit assessment by a nominated ECAI of the corresponding individual issuer; for an individual issuer for which a credit assessment by a nominated ECAI is not available:
 - (i) an institution using the approach referred to in Chapter 3 shall map the internal rating of the individual issuer to one of the external credit assessments;
 - (ii) an institution using the approach referred to in Chapter 2 shall assign $\text{SF}_{k,l}^{\text{Credit}} = 0,54\%$ to that credit reference entity; however, where an institution applies Article 128 to risk weight counterparty credit risk exposures to that individual issuer, $\text{SF}_{k,l}^{\text{Credit}} = 1,6\%$ shall be assigned to that credit reference entity;
- (b) for the credit reference entity k established in accordance with point (b) of paragraph 1:

▼M8

- (i) where a risk position l assigned to the credit reference entity k is a credit index listed on a recognised exchange, $SF_{k,l}^{\text{Credit}}$ shall be mapped to one of the two supervisory factors set out in Table 4 of this paragraph on the basis of the credit quality of the majority of its individual constituents;
- (ii) where a risk position l assigned to the credit reference entity k is not referred to in point (i) of this point, $SF_{k,l}^{\text{Credit}}$ shall be the weighted average of the supervisory factors mapped to each constituent in accordance with the method set out in point (a), where the weights are defined by the proportion of notional of the constituents in that position.

Table 3

Credit quality step	Supervisory factor for single-name transactions
1	0,38 %
2	0,42 %
3	0,54 %
4	1,06 %
5	1,6 %
6	6,0 %

Table 4

Dominant credit quality	Supervisory factor for quoted indices
Investment grade	0,38 %
Non-investment grade	1,06 %

*Article 280d***Equity risk category add-on**

1. For the purposes of paragraph 2, institutions shall establish the relevant equity reference entities of the netting set in accordance with the following:

- (a) there shall be one equity reference entity for each issuer of a reference equity instrument that underlies a single-name transaction allocated to the equity risk category; single-name transactions shall be assigned to the same equity reference entity only where the underlying reference equity instrument of those transactions is issued by the same issuer;
- (b) there shall be one equity reference entity for each group of reference equity instruments or single-name equity derivatives that underlie a multi-name transaction allocated to the equity risk category; multi-names transactions shall be assigned to the same equity reference entity only where the group of underlying reference equity instruments or single-name equity derivatives of those transactions, as applicable, has the same constituents.

▼ M8

2. For the purposes of Article 278, institutions shall calculate the equity risk category add-on for a given netting set as follows:

$$\text{AddOn}^{\text{Equity}} = \sum_j \text{AddOn}_j^{\text{Equity}}$$

where:

$\text{AddOn}^{\text{Equity}}$ = the equity risk category add-on;

j = the index that denotes all the equity risk hedging sets established in accordance with point (d) of Article 277a (1) and Article 277a(2) for the netting set; and

$\text{AddOn}_j^{\text{Equity}}$ = the equity risk category add-on for hedging set j calculated in accordance with paragraph 3.

3. Institutions shall calculate the equity risk category add-on for hedging set j as follows:

▼ C7

$$\text{AddOn}_j^{\text{Equity}} = \epsilon_j \sqrt{\left(\sum_k \rho_k^{\text{Equity}} \cdot \text{AddOn}(\text{Entity}_k) \right)^2 + \sum_k \left(1 - (\rho_k^{\text{Equity}})^2 \right) \cdot (\text{AddOn}(\text{Entity}_k))^2}$$

▼ M8

where:

$\text{AddOn}_j^{\text{Equity}}$ = the equity risk category add-on for hedging set j ;

ϵ_j = the hedging set supervisory factor coefficient of hedging set j determined in accordance with Article 280;

k = the index that denotes the equity reference entities of the netting set established in accordance with paragraph 1;

ρ_k^{Equity} = the correlation factor of the equity reference entity k ; where the equity reference entity k has been established in accordance with point (a) of paragraph 1, $\rho_k^{\text{Equity}} = 50\%$; where the equity reference entity k has been established in accordance with point (b) of paragraph 1, $\rho_k^{\text{Equity}} = 80\%$; and

$\text{AddOn}(\text{Entity}_k)$ = the add-on for the equity reference entity k determined in accordance with paragraph 4.

4. Institutions shall calculate the add-on for the equity reference entity k as follows:

▼ C7

$$\text{AddOn}(\text{Entity}_k) = \text{SF}_k^{\text{Equity}} \cdot \text{EffNot}_k^{\text{Equity}}$$

▼ M8

where:

$\text{AddOn}(\text{Entity}_k)$ = the add-on for the equity reference entity k ;

▼ M8

SF_k^{Equity} = the supervisory factor applicable to the equity reference entity k ; where the equity reference entity k has been established in accordance with point (a) of paragraph 1, $SF_k^{\text{Equity}} = 32\%$; where the equity reference entity k has been established in accordance with point (b) of paragraph 1, $SF_k^{\text{Equity}} = 20\%$; and

$\text{EffNot}_k^{\text{Equity}}$ = the effective notional amount of the equity reference entity k calculated as follows:

$$\text{EffNot}_k^{\text{Equity}} = \sum_{l \in \text{Equity reference entity } k} \text{RiskPosition}_l$$

where:

l = the index that denotes the risk position.

*Article 280e***Commodity risk category add-on**

1. For the purposes of Article 278, institutions shall calculate the commodity risk category add-on for a given netting set as follows:

▼ C7

$$\text{AddOn}^{\text{Com}} = \sum_j \text{AddOn}_j^{\text{Com}}$$

▼ M8

where:

$\text{AddOn}^{\text{Com}}$ = the commodity risk category add-on;

j = the index that denotes the commodity hedging sets established in accordance with point (e) of Article 277a(1) and with Article 277a(2) for the netting set; and

$\text{AddOn}_j^{\text{Com}}$ = the commodity risk category add-on for hedging set j calculated in accordance with paragraph 4.

2. For the purpose of calculating the add-on for a commodity hedging set of a given netting set in accordance with paragraph 4, institutions shall establish the relevant commodity reference types of each hedging set. Commodity derivative transactions shall be assigned to the same commodity reference type only where the underlying commodity instrument of those transactions has the same nature, irrespective of the delivery location and quality of the commodity instrument.

3. By way of derogation from paragraph 2, competent authorities may require an institution which is significantly exposed to the basis risk of different positions sharing the same nature as referred to in paragraph 2 to establish the commodity reference types for those positions using more characteristics than just the nature of the underlying commodity instrument. In such a situation, commodity derivative transactions shall be assigned the same commodity reference type only where they share those characteristics.

▼M8

4. Institutions shall calculate the commodity risk category add-on for hedging set j as follows:

▼C7

$$\text{AddOn}_j^{\text{Com}} = \epsilon_j \sqrt{\left(\rho^{\text{Com}} \cdot \sum_k \text{AddOn}(\text{Type}_k^j) \right)^2 + (1 - (\rho^{\text{Com}})^2) \cdot \sum_k \left(\text{AddOn}(\text{Type}_k^j) \right)^2}$$

▼M8

where:

$\text{AddOn}_j^{\text{Com}}$ = the commodity risk category add-on for hedging set j;

ϵ_j = the hedging set supervisory factor coefficient of hedging set j determined in accordance with Article 280;

ρ^{Com} = the correlation factor of the commodity risk category with a value equal to 40 %;

k = the index that denotes the commodity reference types of the netting set established in accordance with paragraph 2; and

$\text{AddOn}(\text{Type}_k^j)$ = the add-on for the commodity reference type k calculated in accordance with paragraph 5.

5. Institutions shall calculate the add-on for the commodity reference type k as follows:

$$\text{AddOn}(\text{Type}_k^j) = \text{SF}_k^{\text{Com}} \cdot \text{EffNot}_k^{\text{Com}}$$

where:

$\text{AddOn}(\text{Type}_k^j)$ = the add-on for the commodity reference type k;

▼C7

SF_k^{Com} = the supervisory factor applicable to the commodity reference type k;

where the commodity reference type k corresponds to transactions allocated to the hedging set referred to in point (e) of Article 277a(1), excluding transactions concerning electricity, $\text{SF}_k^{\text{Com}} = 18\%$; for transactions concerning electricity, $\text{SF}_k^{\text{Com}} = 40\%$; and

▼M8

$\text{EffNot}_k^{\text{Com}}$ = the effective notional amount of the commodity reference type k calculated as follows:

$$\text{EffNot}_k^{\text{Com}} = \sum_{l \in \text{Commodity reference type k}} \text{RiskPosition}_l$$

where:

l = the index that denotes the risk position.

▼ **M8***Article 280f***Other risks category add-on**

1. For the purposes of Article 278, institutions shall calculate the other risks category add-on for a given netting set as follows:

$$\text{AddOn}^{\text{Other}} = \sum_j \text{AddOn}_j^{\text{Other}}$$

where:

$\text{AddOn}^{\text{Other}}$ = the other risks category add-on;

ϵ_j = the index that denotes the other risk hedging sets established in accordance with point (f) of Article 277a(1) and Article 277a(2) for the netting set; and

$\text{AddOn}_j^{\text{Other}}$ = the other risks category add-on for hedging set j calculated in accordance with paragraph 2.

2. Institutions shall calculate the other risks category add-on for hedging set j as follows:

$$\text{AddOn}_j^{\text{Other}} = \epsilon_j \cdot \text{SF}^{\text{Other}} \cdot |\text{EffNot}_j^{\text{Other}}|$$

where:

$\text{AddOn}_j^{\text{Other}}$ = the other risks category add-on for hedging set j;

ϵ_j = the hedging set supervisory factor coefficient of hedging set j determined in accordance with Article 280; and

SF^{Other} = the supervisory factor for the other risk category with a value equal to 8 %;

$\text{EffNot}_j^{\text{Other}}$ = the effective notional amount of hedging set j calculated as follows:

$$\text{EffNot}_j^{\text{Other}} = \sum_{l \in \text{Hedging set } j} \text{RiskPosition}_l$$

where:

l = the index that denotes the risk position.

Section 4**Simplified standardised approach for counterparty credit risk***Article 281***Calculation of the exposure value**

1. Institutions shall calculate a single exposure value at netting set level in accordance with Section 3, subject to paragraph 2 of this Article.

2. The exposure value of a netting set shall be calculated in accordance with the following requirements:

- (a) institutions shall not apply the treatment referred to in Article 274 (6);

▼M8

- (b) by way of derogation from Article 275(1), for netting sets that are not referred to in Article 275(2), institutions shall calculate the replacement cost in accordance with the following formula:

$$RC = \max\{CMV, 0\}$$

where:

RC = the replacement cost; and

CMV = the current market value.

- (c) by way of derogation from Article 275(2) of this Regulation, for netting sets of transactions: that are traded on a recognised exchange; that are centrally cleared by a central counterparty authorised in accordance with Article 14 of Regulation (EU) No 648/2012 or recognised in accordance with Article 25 of that Regulation; or for which collateral is exchanged bilaterally with the counterparty in accordance with Article 11 of Regulation (EU) No 648/2012, institutions shall calculate the replacement cost in accordance with the following formula:

$$RC = TH + MTA$$

where:

RC = the replacement cost;

TH = the margin threshold applicable to the netting set under the margin agreement below which the institution cannot call for collateral; and

MTA = the minimum transfer amount applicable to the netting set under the margin agreement;

- (d) by way of derogation from Article 275(3), for multiple netting sets that are subject to a margin agreement, institutions shall calculate the replacement cost as the sum of the replacement cost of each individual netting set, calculated in accordance with paragraph 1 as if they were not margined;
- (e) all hedging sets shall be established in accordance with Article 277a(1);
- (f) institutions shall set to 1 the multiplier in the formula that is used to calculate the potential future exposure in Article 278(1), as follows:

$$PFE = \sum_a \text{AddOn}^{(a)}$$

where:

PFE = the potential future exposure; and

AddOn^(a) = the add-on for risk category a;

- (g) by way of derogation from Article 279a(1), for all transactions, institutions shall calculate the supervisory delta as follows:

▼M8

$$\delta = \begin{cases} +1 & \text{where the transaction is a long position in the primary risk driver} \\ -1 & \text{where the transaction is a short position in the primary risk driver} \end{cases}$$

where:

δ = the supervisory delta;

- (h) the formula referred to in point (a) of Article 279b(1) that is used to compute the supervisory duration factor shall read as follows:

$$\text{supervisory duration factor} = E - S$$

where:

E = the period between the end date of a transaction and the reporting date; and

S = the period between the start date of a transaction and the reporting date;

- (i) the maturity factor referred to in Article 279c(1) shall be calculated as follows:
- (i) for transactions included in netting sets referred to in Article 275(1), $MF = 1$;
 - (ii) for transactions included in netting sets referred to in Article 275(2) and (3), $MF = 0,42$;
- (j) the formula referred to in Article 280a(3) that is used to calculate the effective notional amount of hedging set j shall read as follows:

$$\text{EffNot}_j^{\text{IR}} = |D_{j,1}| + |D_{j,2}| + |D_{j,3}|$$

where:

$\text{EffNot}_j^{\text{IR}}$ = the effective notional amount of hedging set j ; and

$D_{j,k}$ = the effective notional amount of bucket k of hedging set j ;

- (k) the formula referred to in Article 280c(3) that is used to calculate the credit risk category add-on for hedging set j shall read as follows:

$$\text{AddOn}_j^{\text{Credit}} = \sum_k |\text{AddOn}(\text{Entity}_k)|$$

where:

$\text{AddOn}_j^{\text{Credit}}$ = the credit risk category add-on for hedging set j ; and

$\text{AddOn}(\text{Entity}_k)$ = the add-on for the credit reference entity k ;

▼M8

- (l) the formula referred to in Article 280d(3) that is used to calculate the equity risk category add-on for hedging set j shall read as follows:

$$\text{AddOn}_j^{\text{Equity}} = \sum_k |\text{AddOn}(\text{Entity}_k)|$$

where:

$\text{AddOn}_j^{\text{Equity}}$ = the equity risk category add-on for hedging set j; and

$\text{AddOn}(\text{Entity}_k)$ = the add-on for the credit reference entity k;

- (m) the formula referred to in Article 280e(4) that is used to calculate the commodity risk category add-on for hedging set j shall read as follows:

$$\text{AddOn}_j^{\text{Com}} = \sum_k |\text{AddOn}(\text{Type}_k^j)|$$

where:

$\text{AddOn}_j^{\text{Com}}$ = the commodity risk category add-on for hedging set j; and

$\text{AddOn}(\text{Type}_k^j)$ = the add-on for the commodity reference type k.

Section 5

Original exposure method

Article 282

Calculation of the exposure value

1. Institutions may calculate a single exposure value for all the transactions within a contractual netting agreement where all the conditions set out in Article 274(1) are met. Otherwise, institutions shall calculate an exposure value separately for each transaction, which shall be treated as its own netting set.
2. The exposure value of a netting set or a transaction shall be the product of 1,4 times the sum of the current replacement cost and the potential future exposure.
3. The current replacement cost referred to in paragraph 2 shall be calculated as follows:
 - (a) for netting sets of transactions: that are traded on a recognised exchange; centrally cleared by a central counterparty authorised in accordance with Article 14 of Regulation (EU) No 648/2012 or recognised in accordance with Article 25 of that Regulation; or for which collateral is exchanged bilaterally with the counterparty in accordance with Article 11 of Regulation (EU) No 648/2012, institutions shall use the following formula:

▼M8

$$RC = TH + MTA$$

where:

RC = the replacement cost;

TH = the margin threshold applicable to the netting set under the margin agreement below which the institution cannot call for collateral; and

MTA = the minimum transfer amount applicable to the netting set under the margin agreement;

- (b) for all other netting sets or individual transactions, institutions shall use the following formula:

$$RC = \max\{CMV, 0\}$$

where:

RC = the replacement cost; and

CMV = the current market value.

In order to calculate the current replacement cost, institutions shall update current market values at least monthly.

4. Institutions shall calculate the potential future exposure referred to in paragraph 2 as follows:

- (a) the potential future exposure of a netting set is the sum of the potential future exposure of all the transactions included in the netting set, calculated in accordance with point (b);
- (b) the potential future exposure of a single transaction is its notional amount multiplied by:
 - (i) the product of 0,5 % and the residual maturity of the transaction expressed in years for interest-rate derivative contracts;
 - (ii) the product of 6 % and the residual maturity of the transaction expressed in years for credit derivative contracts;
 - (iii) 4 % for foreign-exchange derivatives;
 - (iv) 18 % for gold and commodity derivatives other than electricity derivatives;
 - (v) 40 % for electricity derivatives;
 - (vi) 32 % for equity derivatives;

▼M8

- (c) the notional amount referred to in point (b) of this paragraph shall be determined in accordance with Article 279b(2) and (3) for all derivatives listed in that point; in addition, the notional amount of the derivatives referred to in points (b)(iii) to (b)(vi) of this paragraph shall be determined in accordance with points (b) and (c) of Article 279b(1);
- (d) the potential future exposure of netting sets referred to in point (a) of paragraph 3 shall be multiplied by 0,42.

For calculating the potential exposure of interest-rate derivatives and credit derivatives in accordance with points b(i) and (b)(ii), an institution may choose to use the original maturity instead of the residual maturity of the contracts.

▼C2**Section 6****Internal Model Method***Article 283***Permission to use the Internal Model Method**

1. Provided that the competent authorities are satisfied that the requirement in paragraph 2 have been met by an institution, they shall permit that institution to use the Internal Model Method (IMM) to calculate the exposure value for any of the following transactions:

- (a) transactions in Article 273(2)(a);
- (b) transactions in Article 273(2)(b), (c) and (d);
- (c) transactions in Article 273(2)(a) to (d),

Where an institution is permitted to use the IMM to calculate exposure value for any of the transactions mentioned in points (a) to (c) of the first subparagraph, it may also use the IMM for the transactions in Article 273(2)(e).

Notwithstanding the third subparagraph of Article 273(1), an institution may choose not to apply this method to exposures that are immaterial in size and risk. In such case, an institution shall apply one of the methods set out in Sections 3 to 5 to these exposures where the relevant requirements for each approach are met.

2. Competent authorities shall permit institutions to use IMM for the calculations referred to in paragraph 1 only if the institution has demonstrated that it complies with the requirements set out in this Section, and the competent authorities verified that the systems for the management of CCR maintained by the institution are sound and properly implemented.

3. The competent authorities may permit institutions for a limited period to implement the IMM sequentially across different transaction types. During this period of sequential implementation institutions may use the methods set out in Section 3 or Section 5 for transaction type for which they do not use the IMM.

▼ **C2***Article 285***Exposure value for netting sets subject to a margin agreement**

1. If the netting set is subject to a margin agreement and daily mark-to-market valuation, the institution shall calculate Effective EPE as set out in this paragraph. If the model captures the effects of margining when estimating EE, the institution may, subject to the permission of the competent authority, use the model's EE measure directly in the equation in Article 284(5). Competent authorities shall grant such permission only if they verify that the model properly captures the effects of margining when estimating EE. An institution that has not received such permission shall use one of the following Effective EPE measures:

- (a) Effective EPE, calculated without taking into account any collateral held or posted by way of margin plus any collateral that has been posted to the counterparty independent of the daily valuation and margining process or current exposure;
- (b) Effective EPE, calculated as the potential increase in exposure over the margin period of risk, plus the larger of:
 - (i) the current exposure including all collateral currently held or posted, other than collateral called or in dispute;
 - (ii) the largest net exposure, including collateral under the margin agreement, that would not trigger a collateral call. This amount shall reflect all applicable thresholds, minimum transfer amounts, independent amounts and initial margins under the margin agreement.

For the purposes of point (b), institutions shall calculate the add-on as the expected positive change of the mark-to-market value of the transactions during the margin period of risk. Changes in the value of collateral shall be reflected using the Supervisory Volatility Adjustments Approach in accordance with Section 4 of Chapter 4 or the own estimates of volatility adjustments of the Financial Collateral Comprehensive Method, but no collateral payments shall be assumed during the margin period of risk. The margin period of risk is subject to the minimum periods set out in paragraphs 2 to 5.

2. For transactions subject to daily re-margining and mark-to-market valuation, the margin period of risk used for the purpose of modelling the exposure value with margin agreements shall not be less than:

- (a) 5 business days for netting sets consisting only of repurchase transactions, securities or commodities lending or borrowing transactions and margin lending transactions;
- (b) 10 business days for all other netting sets.

3. Points (a) and (b) of paragraph 2 shall be subject to the following exceptions:

- (a) for all netting sets where the number of trades exceeds 5 000 at any point during a quarter, the margin period of risk for the following quarter shall not be less than 20 business days. This exception shall not apply to institutions' trade exposures;

▼ C2

- (b) for netting sets containing one or more trades involving either illiquid collateral, or an OTC derivative that cannot be easily replaced, the margin period of risk shall not be less than 20 business days.

An institution shall determine whether collateral is illiquid or whether OTC derivatives cannot be easily replaced in the context of stressed market conditions, characterised by the absence of continuously active markets where a counterparty would, within two days or fewer, obtain multiple price quotations that would not move the market or represent a price reflecting a market discount (in the case of collateral) or premium (in the case of an OTC derivative).

An institution shall consider whether trades or securities it holds as collateral are concentrated in a particular counterparty and if that counterparty exited the market precipitously whether the institution would be able to replace those trades or securities.

4. If an institution has been involved in more than two margin call disputes on a particular netting set over the immediately preceding two quarters that have lasted longer than the applicable margin period of risk under paragraphs 2 and 3, the institution shall use a margin period of risk that is at least double the period specified in paragraphs 2 and 3 for that netting set for the subsequent two quarters.

5. For re-margining with a periodicity of N days, the margin period of risk shall be at least equal to the period specified in paragraphs 2 and 3, F , plus N days minus one day. That is:

$$\text{Margin Period of Risk} = F + N - 1$$

6. If the internal model includes the effect of margining on changes in the market value of the netting set, an institution shall model collateral, other than cash of the same currency as the exposure itself, jointly with the exposure in its exposure value calculations for OTC derivatives and securities-financing transactions.

▼ M17

7. If an institution is not able to model collateral jointly with the exposure, it shall not recognise in its exposure value calculations for OTC derivatives the effect of collateral other than cash of the same currency as the exposure itself, unless the institution uses the volatility adjustments under the standard Supervisory Volatility Adjustments Approach in accordance with Chapter 4.

7a. If an institution is not able to model collateral jointly with the exposure, it shall not recognise in its exposure value calculations for securities financing transactions the effect of collateral other than cash of the same currency as the exposure itself.

▼ C2

8. An institution using the IMM shall ignore in its models the effect of a reduction of the exposure value due to any clause in a collateral agreement that requires receipt of collateral when counterparty credit quality deteriorates.

▼ **C2***Article 286***Management of CCR — Policies, processes and systems**

1. An institution shall establish and maintain a CCR management framework, consisting of:

- (a) policies, processes and systems to ensure the identification, measurement, management, approval and internal reporting of CCR;
- (b) procedures for ensuring that those policies, processes and systems are complied with.

Those policies, processes and systems shall be conceptually sound, implemented with integrity and documented. The documentation shall include an explanation of the empirical techniques used to measure CCR.

2. The CCR management framework required by paragraph 1 shall take account of market, liquidity, and legal and operational risks that are associated with CCR. In particular, the framework shall ensure that the institution complies with the following principles:

- (a) it does not undertake business with a counterparty without assessing its creditworthiness;
- (b) it takes due account of settlement and pre-settlement credit risk;
- (c) it manages such risks as comprehensively as practicable at the counterparty level by aggregating CCR exposures with other credit exposures and at the firm-wide level.

3. An institution using the IMM shall ensure that its CCR management framework accounts to the satisfaction of the competent authority for the liquidity risks of all of the following:

- (a) potential incoming margin calls in the context of exchanges of variation margin or other margin types, such as initial or independent margin, under adverse market shocks;
- (b) potential incoming calls for the return of excess collateral posted by counterparties;
- (c) calls resulting from a potential downgrade of its own external credit quality assessment.

An institution shall ensure that the nature and horizon of collateral re-use is consistent with its liquidity needs and does not jeopardise its ability to post or return collateral in a timely manner.

4. An institution's management body and senior management shall be actively involved in, and ensure that adequate resources are allocated to, the management of CCR. Senior management shall be aware of the limitations and assumptions of the model used and the impact those limitations and assumptions can have on the reliability of the output through a formal process. Senior management shall be also aware of the uncertainties of the market environment and operational issues and of how these are reflected in the model.